Longitudinal Analysis of Hair Metabolome Dynamics: Temporal Trajectories

Data Analysis Report

February 15, 2026

Abstract

This study investigates the longitudinal metabolic alterations in hair samples across three distinct timepoints: Before, During, and After a specific intervention. Using two complementary Liquid Chromatography-Mass Spectrometry (LC-MS) platforms—HILIC (Hydrophilic Interaction Liquid Chromatography) and HSST3 (High Strength Silica T3 for hydrophobic compounds)—we profiled the hair metabolome of a patient cohort ($n = 7$). Statistical analysis employed non-parametric Friedman tests for global variance and Wilcoxon Signed-Rank tests for pairwise comparisons ($p < 0.05$). We identified a consistent "Return to Baseline" (RTB) phenotype in a subset of metabolites (12 in HILIC, 13 in HSST3, see Appendix A), characterized by a significant deviation during the 'During' phase followed by detailed recovery.

1 Methodology

1.1 Study Design and Data Preprocessing

Hair samples were collected from a cohort of patients at three timepoints:

- **Before**: Baseline state (Samples 6-7).
- **During**: Intervention/Treatment phase (Sample 5).
- **After**: Post-intervention phase (Samples 1-4).

Data was acquired using two LC-MS methods:

- **HILIC (Pos)**: detection of 1196 polar metabolic features.
- **HSST3 (Pos)**: detection of 2610 hydrophobic metabolic features.

1.2 Data Normalization

To account for inter-patient variability, a **Patient-Specific Baseline Normalization** is applied:

1. **Baseline Calculation**: For each patient, the mean abundance of "Before" samples is calculated.
2. **Log-Transformation**: The abundance values are transformed to **Log2 Fold Changes (Log2FC)** relative to this baseline:

$$\text{Log2FC} = \log_2 \left(\frac{\text{Value} + \epsilon}{\text{Baseline} + \epsilon} \right) \tag{1}$$

(where $\epsilon = 10^{-6}$ to prevent division by zero)

This ensures that all subsequent analysis measures *change* relative to the patient's own starting point.

1.3 Statistical Framework

Due to the small sample size and potential non-normality of biological data, rigorous non-parametric tests were employed.

1.3.1 Friedman Test (Non-parametric Repeated Measures)

- **Comparison**: Global test across all 3 periods (Before, During, After).
- **Data Included**: Aggregated means per patient for each period.
- **Method**: Extension of the Sign Test for > 2 groups. Ranks each patient's values across periods.
- **Requirement**: Must have data for all 3 periods for a patient to include them (Complete Cases).
- **Nuance**: Appropriately handles the repeated measures design (pairing), unlike independent group tests (e.g., Kruskal-Wallis).

1.3.2 Wilcoxon Signed-Rank Test (Paired)

- **Comparison**: Specific Pairwise Comparisons (e.g., Before vs. During).
- **Data Included**:
  1. First, calculates the **mean** Log2FC for each patient in each period.
  2. Then, only includes patients who have data for **BOTH** timepoints being compared.
- **Method**: Calculates differences between paired means, ranks the absolute differences.
- **Nuance**: Most conservative test. It correctly handles the "paired" nature of the study (same patient measured over time).

1.3.3 Significance Threshold

A raw p-value threshold of $p < 0.05$ was used for identifying candidate features. Feature Discovery Rate (FDR) correction was purposefully omitted to maximize sensitivity for candidate generation in this exploratory phase.

1.4 Feature Selection Criteria

"Return to Baseline" (RTB) Pattern: We defined a specific phenotype of interest representing metabolites that respond to the condition/treatment and then recover. Features were classified as RTB if they met:

1. $P_{Wilcoxon}$(Before vs During) < 0.05 (Significant Shift)
2. $P_{Wilcoxon}$(During vs After) < 0.05 (Significant Recovery)
3. $P_{Wilcoxon}$(Before vs After) ≥ 0.05 (Return to pre-state levels)

2 Results

2.1 Principal Component Analysis (PCA)

To visualize the overall structure of the metabolic data, we performed Principal Component Analysis (PCA) on the normalized feature abundance profiles. The 2D PCA plots (Figure 1) illustrate the distribution of samples across the first two principal components.

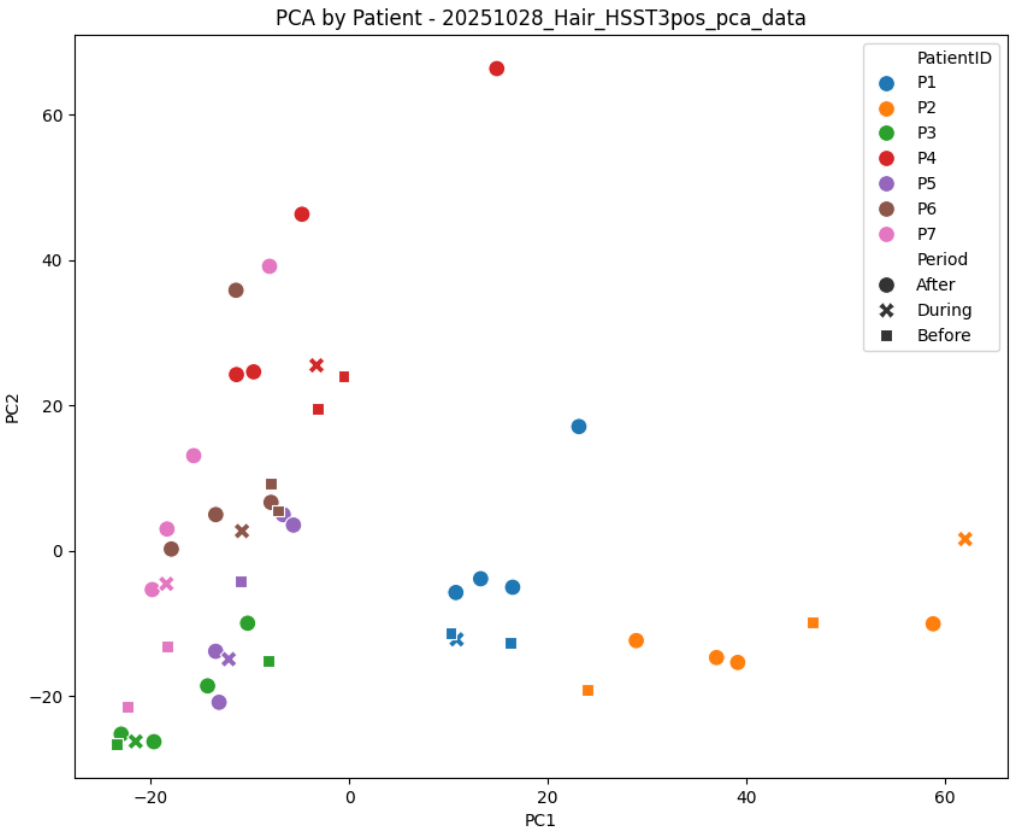


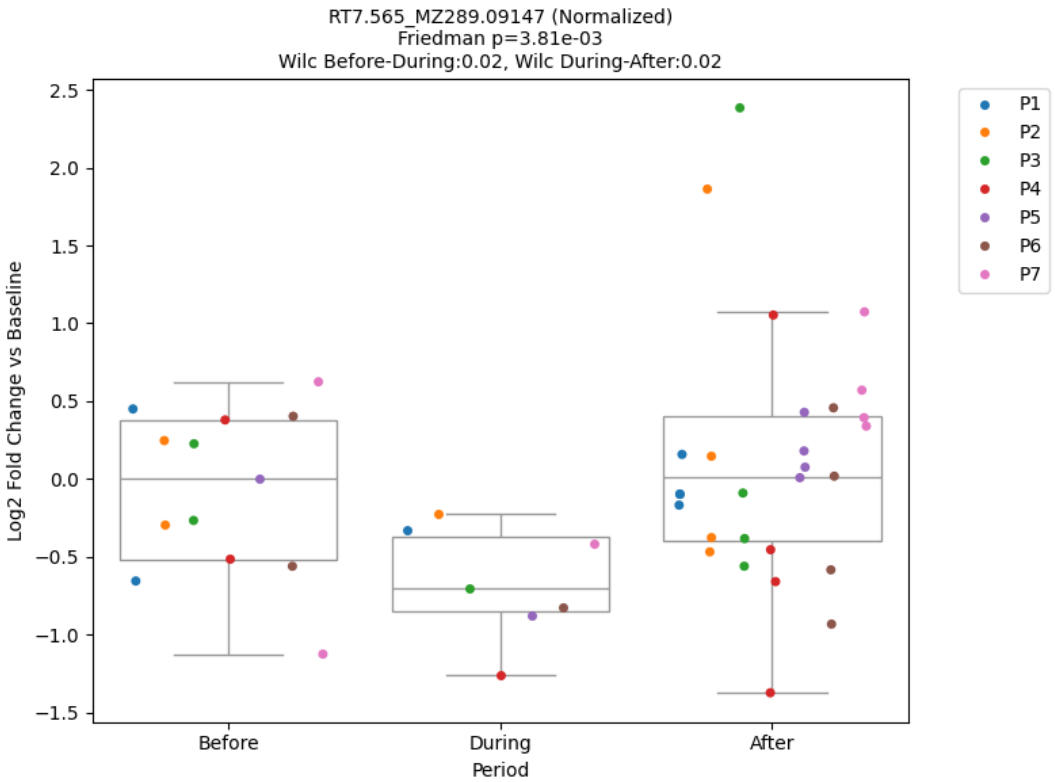
Figure 1: Principal Component Analysis (PCA) of HSST3 Data. Samples are colored by Patient ID to visualize inter-individual variability.

2.2 Global Metabolic Alterations

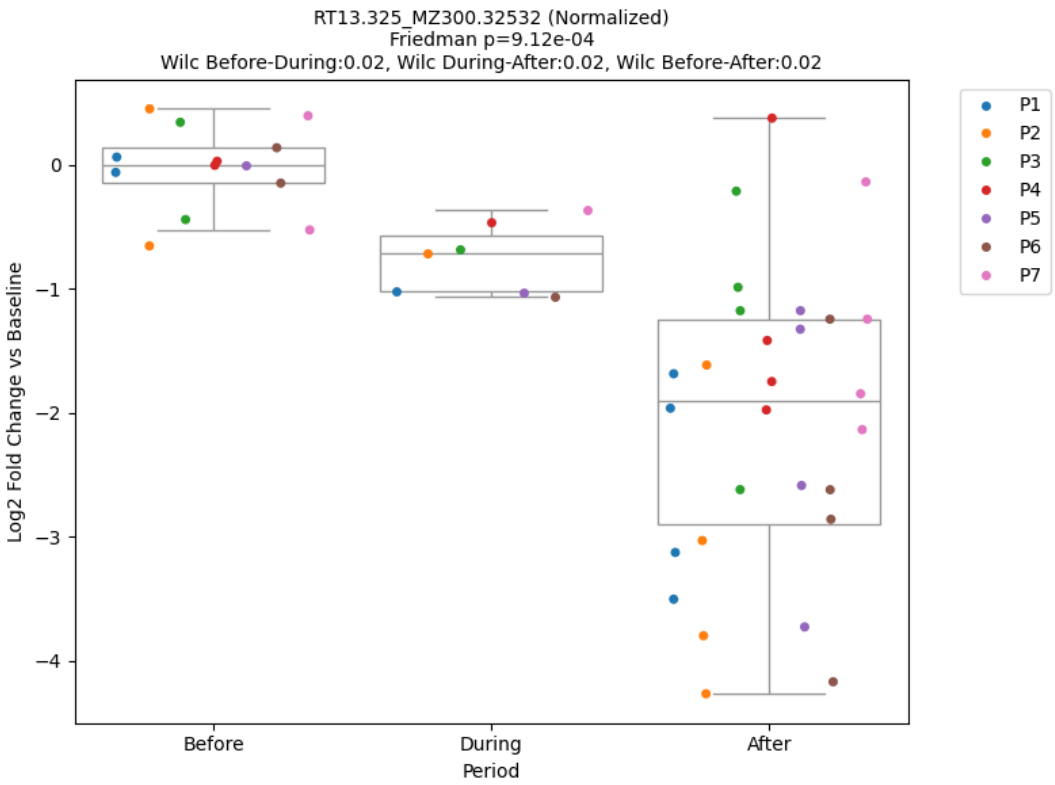
The Friedman test revealed potential global shifts in the metabolome. Both platforms showed a comparable proportion of significantly altered features ($p < 0.05$):

- **HILIC**: 164 features (13.7% vs 5% expected from random data)
- **HSST3**: 352 features (13.5% vs 5% expected from random data)

Figure 2 illustrates the top discriminating features ranked by Friedman p-value.



(a) HILIC Top 1



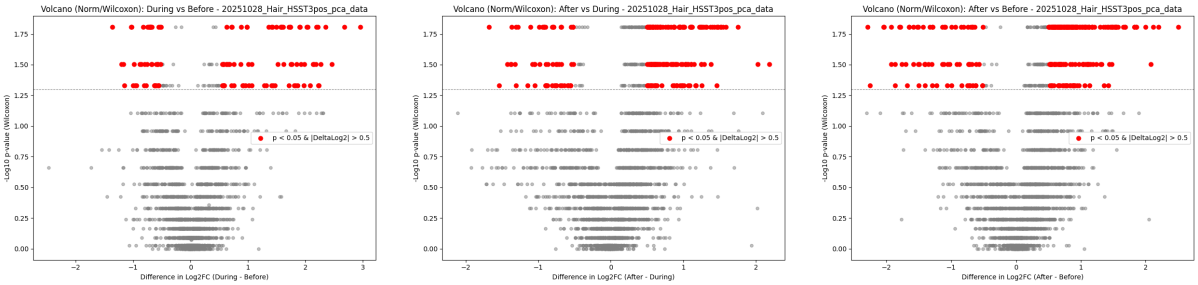
(b) HSST3 Top 1

Figure 2: Top Differentiating Metabolites (Friedman Test). The test calculates differentiation based on patient means per period. Boxplots of Log2 Fold Change across timepoints for representative top features.

2.3 Pairwise Differential Abundance and Temporal Drift

Pairwise Wilcoxon tests dissected the specific timing of metabolic changes.

- **Before vs. During**: ∼3-6% of features showed significant immediate deviation (37 in HILIC, 148 in HSST3, vs 5% expected).
- **During vs. After**: A larger fraction changed between these phases (10.5% in HILIC, 12.1% in HSST3, vs 5% expected).
- **Before vs. After (Long-term)**: The largest proportion of changes occurred here (12.9% in HILIC, 13.9% in HSST3, vs 5% expected). This suggests a substantial "drift" or long-term effect where the metabolome does not return to its original state for many features.



(a) Before vs During (b) During vs After (c) Before vs After

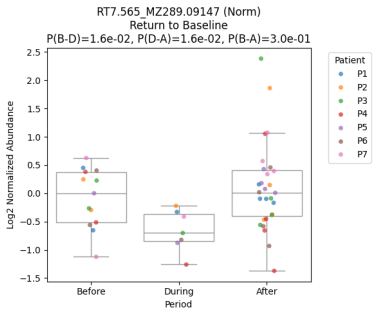
Figure 3: Volcano Plots (HSST3). Red points indicate significant features ($p < 0.05$ and $|\text{Log2FC}| > 0.5$).

2.4 "Return to Baseline" Trajectories

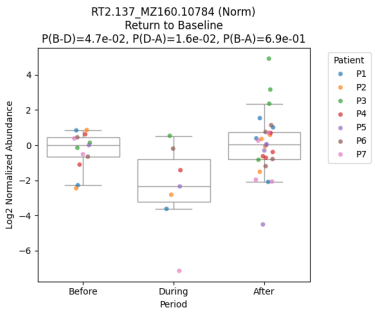
We identified a select subset of metabolites that exhibited a distinct trajectory: a significant excursion during the 'During' phase followed by a complete recovery to baseline levels. This pattern is indicative of a direct, reversible response to the intervention.

- **Identified Candidates**: 12 features in HILIC (Table 1) and 13 in HSST3 (Table 2) met the strict RTB criteria.

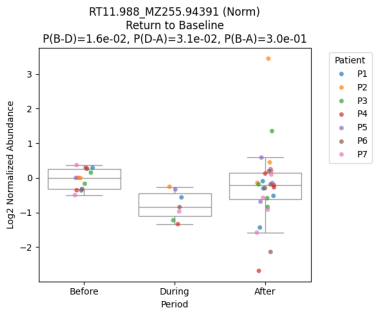
Figure 4 displays the trajectories of these candidates. Individual patient data points are colored to show consistent intra-subject trends.



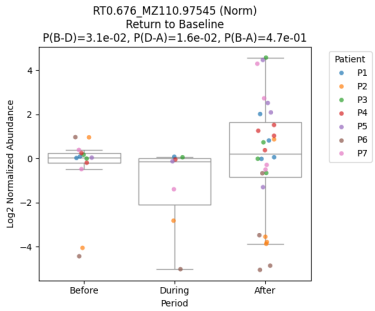
(a) HILIC: RT7.565_MZ289.09147



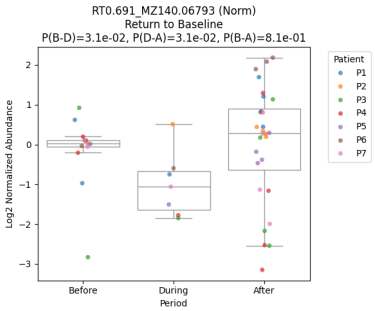
(b) HILIC: RT2.137_MZ160.10784



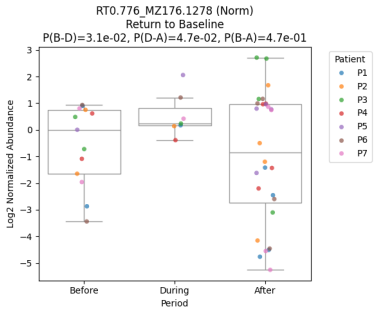
(c) HILIC: RT11.988_MZ255.94391



(d) HSST3: RT0.676_MZ110.97545



(e) HSST3: RT0.691_MZ140.06793



(f) HSST3: RT0.776_MZ176.1278

Figure 4: Return to Baseline Profiles. Boxplots showing the distribution of fold changes. Individual patient values (colored dots) demonstrate the consistency of the 'V-shape' or 'Inverted V-shape' recovery pattern.

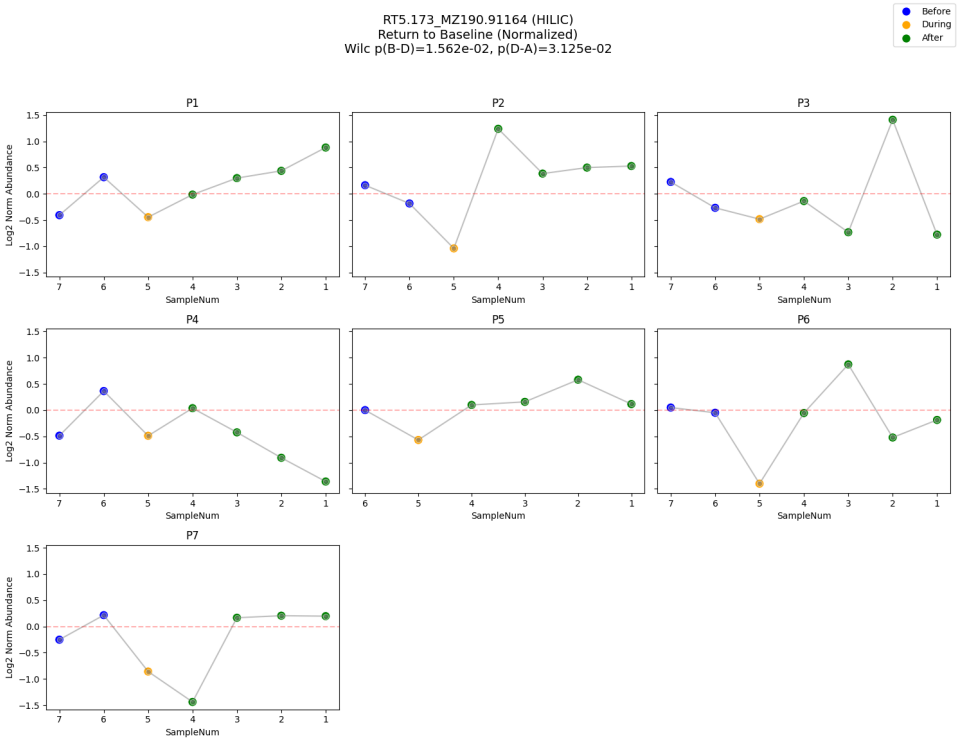


Figure 5: Detailed Timeline of a Return-to-Baseline Feature (RT 5.173, MZ 190.91). This plot illustrates the individual patient trajectories normalized to baseline, highlighting the reversible response to the intervention.

3 Discussion

The analysis has identified specific metabolic features with distinct temporal patterns.

A Compound Lists

A.1 HILIC - Return to Baseline (RTB) Candidates

The following 12 features were identified in the HILIC dataset as showing a significant "Return to Baseline" pattern:

Compound Identifier (RT_MZ)	**Wilcoxon p (Before-During)**
RT7.565_MZ289.09147	0.0156
RT7.56_MZ433.0924	0.0156
RT11.01_MZ318.93698	0.0156
RT11.988_MZ255.94391	0.0156
RT5.743_MZ216.92239	0.0156
RT5.173_MZ190.91164	0.0156
RT5.103_MZ494.80901	0.0156
RT6.086_MZ156.07416	0.0312
RT2.137_MZ160.10784	0.0469
RT7.572_MZ423.06362	0.0469
RT7.561_MZ450.08254	0.0469
RT7.547_MZ472.10331	0.0469

Table 1: HILIC features exhibiting Return to Baseline phenotype.

A.2 HSST3 - Return to Baseline (RTB) Candidates

The following 13 features were identified in the HSST3 dataset as showing a significant "Return to Baseline" pattern:

Compound Identifier (RT_MZ)	**Wilcoxon p (Before-During)**
RT17.085_MZ165.99057	0.0156
RT5.994_MZ472.27428	0.0156
RT6.543_MZ446.27696	0.0156
RT0.676_MZ110.97545	0.0312
RT0.78_MZ146.10526	0.0312
RT7.99_MZ125.09603	0.0312
RT15.601_MZ590.42472	0.0312
RT14.365_MZ299.21868	0.0312
RT17.078_MZ207.01699	0.0312
RT0.691_MZ140.06793	0.0312
RT0.776_MZ176.1278	0.0312
RT12.704_MZ262.2007	0.0312
RT17.079_MZ166.9939	0.0469

Table 2: HSST3 features exhibiting Return to Baseline phenotype.

B HILIC Significant Feature Lists

B.1 HILIC - Friedman (Global) (164 features)

Compound	**p-val**	**Compound**	**p-val**	**Compound**	**p-val**
RT7.565_MZ289.09147	0.0038	RT5.235_MZ178.086	0.0038	RT4.052_MZ126.06609	0.0038
RT1.766_MZ140.04525	0.0038	RT8.796_MZ199.08412	0.0038	RT8.797_MZ183.11021	0.0038
RT8.789_MZ233.16064	0.0051	RT6.977_MZ189.12322	0.0051	RT5.291_MZ226.15476	0.0051
RT11.01_MZ318.93698	0.0051	RT7.56_MZ433.0924	0.0051	RT3.929_MZ140.08166	0.0051
RT1.318_MZ245.13802	0.0051	RT2.712_MZ137.04561	0.0051	RT11.988_MZ255.94391	0.0058

Compound	p-val	Compound	p-val	Compound	p-val
RT5.161_MZ130.15891	0.0058	RT7.201_MZ112.08682	0.0119	RT2.46_MZ121.03945	0.0119
RT1.089_MZ439.35611	0.0119	RT7.206_MZ112.0868	0.0119	RT1.199_MZ354.40867	0.0119
RT1.715_MZ221.1017	0.0119	RT1.434_MZ315.26382	0.0119	RT5.668_MZ282.11946	0.0119
RT11.945_MZ273.95436	0.0119	RT7.561_MZ450.08254	0.0183	RT5.839_MZ294.93862	0.0183
RT5.291_MZ338.14316	0.0183	RT5.838_MZ158.96392	0.0183	RT2.521_MZ184.16939	0.0183
RT11.928_MZ143.99662	0.0183	RT5.743_MZ216.92239	0.0183	RT7.197_MZ120.06538	0.0183
RT1.43_MZ371.32646	0.0183	RT1.429_MZ476.30585	0.0183	RT2.526_MZ287.23275	0.0183
RT1.415_MZ316.28405	0.0183	RT5.352_MZ295.13167	0.0183	RT7.682_MZ166.05301	0.0183
RT7.19_MZ300.004	0.0183	RT1.186_MZ229.14293	0.0183	RT4.948_MZ176.06558	0.0183
RT3.609_MZ235.1187	0.0183	RT5.317_MZ233.0506	0.0183	RT4.972_MZ131.11775	0.0183
RT9.495_MZ219.0826	0.0183	RT7.816_MZ267.10852	0.0183	RT7.195_MZ164.02923	0.0183
RT5.188_MZ291.86317	0.0183	RT4.428_MZ188.0704	0.0183	RT4.671_MZ176.06554	0.0183
RT8.371_MZ168.07652	0.0183	RT7.572_MZ423.06362	0.0183	RT5.74_MZ298.02465	0.0183
RT3.911_MZ235.11871	0.0183	RT6.086_MZ156.07416	0.0183	RT8.41_MZ343.12311	0.0183
RT2.137_MZ160.10784	0.0183	RT5.756_MZ118.08607	0.0211	RT5.3_MZ217.11803	0.0211
RT7.22_MZ219.07131	0.0211	RT2.488_MZ417.2254	0.0211	RT4.646_MZ88.07557	0.0211
RT5.122_MZ329.08545	0.0211	RT5.389_MZ265.11152	0.0211	RT2.487_MZ418.22176	0.0211
RT3.906_MZ154.09744	0.0211	RT5.306_MZ141.06571	0.0211	RT3.872_MZ269.08778	0.0211
RT4.115_MZ229.11802	0.0211	RT5.103_MZ494.80901	0.0211	RT1.961_MZ182.09224	0.0211
RT7.227_MZ175.1075	0.0211	RT1.155_MZ372.24095	0.0211	RT8.566_MZ203.15001	0.0211
RT7.192_MZ305.06929	0.0211	RT11.929_MZ167.01261	0.0211	RT10.976_MZ192.06148	0.0211
RT4.652_MZ86.0963	0.0211	RT5.305_MZ227.04265	0.0211	RT4.435_MZ205.09697	0.0211
RT5.173_MZ190.91164	0.0211	RT1.048_MZ423.36145	0.0281	RT1.397_MZ234.16956	0.0281
RT8.403_MZ495.20755	0.0281	RT1.422_MZ574.34245	0.0281	RT5.314_MZ328.1599	0.0281
RT1.434_MZ648.37915	0.0281	RT1.086_MZ364.34158	0.0281	RT1.346_MZ428.37282	0.0281
RT1.715_MZ203.09115	0.0281	RT4.921_MZ254.1609	0.0281	RT1.428_MZ608.38427	0.0281
RT3.883_MZ137.04561	0.0281	RT1.728_MZ243.08356	0.0281	RT1.427_MZ343.29515	0.0281
RT3.624_MZ148.1137	0.0281	RT1.36_MZ300.32568	0.0281	RT1.249_MZ613.65938	0.0281
RT11.913_MZ171.0054	0.0281	RT1.78_MZ262.16464	0.0281	RT1.091_MZ401.27532	0.0281
RT1.079_MZ281.24704	0.0498	RT1.115_MZ480.4251	0.0498	RT1.087_MZ421.34548	0.0498
RT5.307_MZ138.05478	0.0498	RT1.117_MZ413.3042	0.0498	RT6.599_MZ104.07045	0.0498
RT1.238_MZ641.69076	0.0498	RT8.402_MZ539.17139	0.0498	RT8.395_MZ381.07896	0.0498
RT1.419_MZ444.3678	0.0498	RT4.676_MZ254.16098	0.0498	RT1.395_MZ327.33648	0.0498
RT7.561_MZ343.12312	0.0498	RT1.875_MZ162.11226	0.0498	RT1.37_MZ385.37828	0.0498
RT1.416_MZ130.15881	0.0498	RT7.783_MZ492.12953	0.0498	RT2.106_MZ315.26405	0.0498
RT1.37_MZ355.36764	0.0498	RT1.511_MZ284.33061	0.0498	RT1.355_MZ402.35719	0.0498
RT1.088_MZ255.23146	0.0498	RT1.25_MZ284.33081	0.0498	RT3.918_MZ152.05652	0.0498
RT1.182_MZ263.20013	0.0498	RT1.038_MZ369.35096	0.0498	RT1.128_MZ455.35102	0.0498
RT2.512_MZ124.08682	0.0498	RT7.562_MZ325.11258	0.0498	RT4.53_MZ166.08612	0.0498
RT5.193_MZ578.77034	0.0498	RT4.61_MZ183.96487	0.0498	RT1.16_MZ416.26701	0.0498
RT2.636_MZ161.12826	0.0498	RT1.421_MZ240.09937	0.0498	RT2.434_MZ134.08103	0.0498
RT2.57_MZ120.05546	0.0498	RT10.575_MZ142.93837	0.0498	RT2.502_MZ491.26215	0.0498
RT5.311_MZ548.29079	0.0498	RT2.445_MZ183.05691	0.0498	RT5.724_MZ234.14456	0.0498
RT1.043_MZ441.372	0.0498	RT1.899_MZ345.3019	0.0498	RT10.584_MZ278.86146	0.0498
RT1.416_MZ490.3214	0.0498	RT1.037_MZ413.37309	0.0498	RT1.315_MZ298.30998	0.0498
RT7.787_MZ577.10698	0.0498	RT4.616_MZ183.96486	0.0498	RT1.762_MZ371.32648	0.0498
RT4.056_MZ126.066	0.0498	RT4.966_MZ182.10207	0.0498	RT1.068_MZ295.22627	0.0498
RT5.267_MZ130.08605	0.0498	RT1.078_MZ502.3729	0.0498	RT4.597_MZ164.9203	0.0498
RT5.693_MZ226.0448	0.0498	RT11.916_MZ387.89794	0.0498	RT2.535_MZ528.33738	0.0498

Compound	p-val	Compound	p-val	Compound	p-val
RT10.578_MZ382.83673	0.0498	RT1.11_MZ347.25738	0.0498		

B.2 HILIC - Before vs During (37 features)

Compound	p-val	Compound	p-val	Compound	p-val
RT7.565_MZ289.09147	0.0156	RT1.318_MZ245.13802	0.0156	RT7.56_MZ433.0924	0.0156
RT11.01_MZ318.93698	0.0156	RT11.988_MZ255.94391	0.0156	RT1.199_MZ354.40867	0.0156
RT1.089_MZ439.35611	0.0156	RT5.188_MZ291.86317	0.0156	RT5.103_MZ494.80901	0.0156
RT1.092_MZ356.31533	0.0156	RT5.173_MZ190.91164	0.0156	RT5.743_MZ216.92239	0.0156
RT1.315_MZ298.30998	0.0312	RT2.512_MZ124.08682	0.0312	RT1.514_MZ256.29957	0.0312
RT1.301_MZ193.14312	0.0312	RT5.193_MZ578.77034	0.0312	RT1.398_MZ328.35673	0.0312
RT1.37_MZ385.37828	0.0312	RT10.578_MZ382.83673	0.0312	RT1.346_MZ428.37282	0.0312
RT1.091_MZ401.27532	0.0312	RT6.086_MZ156.07416	0.0312	RT2.434_MZ134.08103	0.0312
RT1.168_MZ366.29965	0.0469	RT1.36_MZ300.32568	0.0469	RT7.561_MZ450.08254	0.0469
RT1.128_MZ455.35102	0.0469	RT10.578_MZ398.81062	0.0469	RT7.572_MZ423.06362	0.0469
RT7.547_MZ472.10331	0.0469	RT11.029_MZ204.9567	0.0469	RT2.137_MZ160.10784	0.0469
RT1.101_MZ258.24237	0.0469	RT1.715_MZ221.1017	0.0469	RT1.427_MZ256.29954	0.0469
RT1.095_MZ81.04459	0.0469				

B.3 HILIC - During vs After (126 features)

Compound	p-val	Compound	p-val	Compound	p-val
RT7.565_MZ289.09147	0.0156	RT1.429_MZ476.30585	0.0156	RT1.415_MZ316.28405	0.0156
RT11.929_MZ167.01261	0.0156	RT10.976_MZ192.06148	0.0156	RT1.961_MZ182.09224	0.0156
RT3.872_MZ269.08778	0.0156	RT3.906_MZ154.09744	0.0156	RT2.488_MZ417.2254	0.0156
RT1.091_MZ590.42502	0.0156	RT2.487_MZ418.22176	0.0156	RT3.624_MZ148.1137	0.0156
RT3.883_MZ137.04561	0.0156	RT5.314_MZ328.1599	0.0156	RT1.422_MZ574.34245	0.0156
RT8.403_MZ495.20755	0.0156	RT1.397_MZ234.16956	0.0156	RT2.636_MZ161.12826	0.0156
RT1.762_MZ371.32648	0.0156	RT8.402_MZ539.17139	0.0156	RT7.561_MZ343.12312	0.0156
RT7.562_MZ325.11258	0.0156	RT2.106_MZ315.26405	0.0156	RT3.918_MZ152.05652	0.0156
RT1.272_MZ288.18012	0.0156	RT2.507_MZ391.22901	0.0156	RT11.937_MZ474.83279	0.0156
RT5.743_MZ242.92512	0.0156	RT1.436_MZ736.43146	0.0156	RT2.506_MZ392.2254	0.0156
RT11.928_MZ143.99662	0.0156	RT5.838_MZ158.96392	0.0156	RT4.646_MZ88.07557	0.0156
RT2.503_MZ490.2657	0.0156	RT11.01_MZ318.93698	0.0156	RT7.56_MZ433.0924	0.0156
RT5.291_MZ338.14316	0.0156	RT3.929_MZ140.08166	0.0156	RT8.789_MZ233.16064	0.0156
RT5.161_MZ130.15891	0.0156	RT11.945_MZ273.95436	0.0156	RT2.712_MZ137.04561	0.0156
RT8.797_MZ183.11021	0.0156	RT7.206_MZ112.0868	0.0156	RT5.291_MZ226.15476	0.0156
RT8.796_MZ199.08412	0.0156	RT7.201_MZ112.08682	0.0156	RT5.317_MZ233.0506	0.0156
RT6.977_MZ189.12322	0.0156	RT4.052_MZ126.06609	0.0156	RT2.137_MZ160.10784	0.0156
RT7.572_MZ423.06362	0.0156	RT3.911_MZ235.11871	0.0156	RT5.839_MZ294.93862	0.0156
RT6.086_MZ156.07416	0.0156	RT7.561_MZ450.08254	0.0156	RT5.235_MZ178.086	0.0156
RT9.495_MZ219.0826	0.0156	RT1.766_MZ140.04525	0.0156	RT7.547_MZ472.10331	0.0312
RT1.086_MZ546.39895	0.0312	RT1.875_MZ162.11226	0.0312	RT1.546_MZ287.2327	0.0312
RT2.508_MZ390.2327	0.0312	RT1.425_MZ423.28818	0.0312	RT3.886_MZ722.97701	0.0312
RT1.396_MZ512.41483	0.0312	RT2.383_MZ83.08543	0.0312	RT1.422_MZ198.1274	0.0312
RT1.423_MZ198.12741	0.0312	RT1.16_MZ416.26701	0.0312	RT1.517_MZ363.22662	0.0312

Compound	p-val	Compound	p-val	Compound	p-val
RT2.521_MZ184.16939	0.0312	RT4.53_MZ166.08612	0.0312	RT5.737_MZ270.97735	0.0312
RT3.609_MZ235.1187	0.0312	RT5.173_MZ190.91164	0.0312	RT8.41_MZ343.12311	0.0312
RT5.306_MZ141.06571	0.0312	RT5.389_MZ265.11152	0.0312	RT11.913_MZ171.0054	0.0312
RT1.36_MZ300.32568	0.0312	RT8.395_MZ381.07896	0.0312	RT2.736_MZ121.06466	0.0312
RT1.772_MZ520.33216	0.0312	RT11.988_MZ255.94391	0.0312	RT5.724_MZ234.14456	0.0312
RT3.939_MZ178.0972	0.0312	RT4.056_MZ126.066	0.0312	RT4.966_MZ182.10207	0.0312
RT1.09_MZ416.26685	0.0312	RT1.029_MZ83.06022	0.0469	RT10.976_MZ200.97278	0.0469
RT1.429_MZ520.33199	0.0469	RT5.313_MZ372.18605	0.0469	RT1.395_MZ326.34107	0.0469
RT5.28_MZ316.16125	0.0469	RT2.096_MZ259.1286	0.0469	RT2.218_MZ100.11199	0.0469
RT2.019_MZ146.08097	0.0469	RT1.899_MZ345.3019	0.0469	RT1.46_MZ364.21048	0.0469
RT5.668_MZ282.11946	0.0469	RT1.434_MZ315.26382	0.0469	RT1.089_MZ439.35611	0.0469
RT5.743_MZ216.92239	0.0469	RT5.352_MZ295.13167	0.0469	RT5.103_MZ494.80901	0.0469
RT2.502_MZ491.26215	0.0469	RT1.4_MZ272.25779	0.0469	RT1.068_MZ295.22627	0.0469
RT1.078_MZ502.3729	0.0469	RT5.159_MZ144.10176	0.0469	RT4.597_MZ164.9203	0.0469
RT5.307_MZ138.05478	0.0469	RT1.355_MZ402.35719	0.0469	RT1.11_MZ347.25738	0.0469
RT3.639_MZ194.10204	0.0469	RT2.529_MZ285.13431	0.0469	RT5.743_MZ256.12646	0.0469
RT5.306_MZ249.02455	0.0469	RT11.808_MZ142.97617	0.0469	RT1.555_MZ182.09219	0.0469
RT11.932_MZ214.9173	0.0469	RT2.535_MZ528.33738	0.0469	RT1.156_MZ452.39356	0.0469

B.4 HILIC - Before vs After (154 features)

Compound	p-val	Compound	p-val	Compound	p-val
RT5.235_MZ178.086	0.0156	RT4.115_MZ229.11802	0.0156	RT5.389_MZ265.11152	0.0156
RT5.122_MZ329.08545	0.0156	RT5.306_MZ141.06571	0.0156	RT7.22_MZ219.07131	0.0156
RT5.3_MZ217.11803	0.0156	RT1.78_MZ262.16464	0.0156	RT5.756_MZ118.08607	0.0156
RT4.435_MZ205.09697	0.0156	RT5.305_MZ227.04265	0.0156	RT4.652_MZ86.0963	0.0156
RT7.192_MZ305.06929	0.0156	RT8.566_MZ203.15001	0.0156	RT1.155_MZ372.24095	0.0156
RT2.557_MZ337.18216	0.0156	RT1.728_MZ243.08356	0.0156	RT9.86_MZ177.06084	0.0156
RT1.715_MZ203.09115	0.0156	RT7.796_MZ293.01191	0.0156	RT7.792_MZ106.04977	0.0156
RT9.78_MZ191.07643	0.0156	RT7.795_MZ165.98746	0.0156	RT9.786_MZ147.11263	0.0156
RT10.981_MZ250.94958	0.0156	RT7.795_MZ150.01362	0.0156	RT5.312_MZ335.0736	0.0156
RT2.801_MZ99.09156	0.0156	RT5.302_MZ231.04878	0.0156	RT5.301_MZ437.14159	0.0156
RT1.096_MZ263.19998	0.0156	RT3.357_MZ202.1799	0.0156	RT1.086_MZ364.34158	0.0156
RT1.048_MZ423.36145	0.0156	RT7.227_MZ175.1075	0.0156	RT7.682_MZ166.05301	0.0156
RT7.794_MZ277.03803	0.0156	RT2.526_MZ287.23275	0.0156	RT5.291_MZ226.15476	0.0156
RT6.977_MZ189.12322	0.0156	RT8.789_MZ233.16064	0.0156	RT5.668_MZ282.11946	0.0156
RT1.434_MZ315.26382	0.0156	RT1.715_MZ221.1017	0.0156	RT5.352_MZ295.13167	0.0156
RT2.712_MZ137.04561	0.0156	RT8.797_MZ183.11021	0.0156	RT8.796_MZ199.08412	0.0156
RT2.46_MZ121.03945	0.0156	RT1.766_MZ140.04525	0.0156	RT3.929_MZ140.08166	0.0156
RT4.671_MZ176.06554	0.0156	RT7.816_MZ267.10852	0.0156	RT4.972_MZ131.11775	0.0156
RT4.428_MZ188.0704	0.0156	RT7.197_MZ120.06538	0.0156	RT7.19_MZ300.004	0.0156
RT2.521_MZ184.16939	0.0156	RT1.43_MZ371.32646	0.0156	RT7.195_MZ164.02923	0.0156
RT3.609_MZ235.1187	0.0156	RT1.186_MZ229.14293	0.0156	RT8.41_MZ343.12311	0.0156
RT5.74_MZ298.02465	0.0156	RT4.948_MZ176.06558	0.0156	RT8.371_MZ168.07652	0.0156
RT4.052_MZ126.06609	0.0156	RT1.318_MZ245.13802	0.0156	RT1.093_MZ488.32442	0.0312
RT2.444_MZ182.06054	0.0312	RT7.793_MZ345.02535	0.0312	RT1.085_MZ269.21068	0.0312
RT2.859_MZ147.07625	0.0312	RT5.667_MZ182.08098	0.0312	RT1.123_MZ391.34109	0.0312
RT5.311_MZ208.06358	0.0312	RT1.555_MZ182.09219	0.0312	RT2.36_MZ170.09224	0.0312

Compound	p-val	Compound	p-val	Compound	p-val
RT2.494_MZ416.22907	0.0312	RT5.206_MZ164.07033	0.0312	RT1.2_MZ209.15327	0.0312
RT1.176_MZ400.27214	0.0312	RT7.104_MZ232.1515	0.0312	RT4.251_MZ342.21194	0.0312
RT2.438_MZ123.0915	0.0312	RT5.297_MZ204.1228	0.0312	RT5.406_MZ115.08645	0.0312
RT5.314_MZ157.09697	0.0312	RT1.261_MZ334.22179	0.0312	RT7.794_MZ362.01548	0.0312
RT7.796_MZ489.03995	0.0312	RT1.421_MZ371.32641	0.0312	RT10.995_MZ124.08669	0.0312
RT5.489_MZ336.10662	0.0312	RT1.115_MZ480.4251	0.0312	RT7.787_MZ577.10698	0.0312
RT5.267_MZ130.08605	0.0312	RT1.091_MZ401.27532	0.0312	RT1.36_MZ300.32568	0.0312
RT3.906_MZ154.09744	0.0312	RT5.693_MZ226.0448	0.0312	RT1.961_MZ182.09224	0.0312
RT2.445_MZ183.05691	0.0312	RT4.53_MZ166.08612	0.0312	RT6.599_MZ104.07045	0.0312
RT1.087_MZ421.34548	0.0312	RT4.056_MZ126.066	0.0312	RT5.743_MZ256.12646	0.0312
RT7.783_MZ492.12953	0.0312	RT7.201_MZ112.08682	0.0312	RT1.355_MZ402.35719	0.0312
RT2.57_MZ120.05546	0.0312	RT1.043_MZ441.372	0.0312	RT1.25_MZ284.33081	0.0312
RT1.182_MZ263.20013	0.0312	RT2.502_MZ491.26215	0.0312	RT1.11_MZ347.25738	0.0312
RT5.724_MZ234.14456	0.0312	RT1.089_MZ439.35611	0.0312	RT7.206_MZ112.0868	0.0312
RT3.911_MZ235.11871	0.0312	RT2.686_MZ201.12313	0.0469	RT5.307_MZ233.14943	0.0469
RT1.429_MZ476.30585	0.0469	RT4.921_MZ254.1609	0.0469	RT1.346_MZ428.37282	0.0469
RT1.02_MZ420.33129	0.0469	RT1.099_MZ576.37658	0.0469	RT4.108_MZ124.08679	0.0469
RT9.787_MZ147.11259	0.0469	RT1.946_MZ201.12312	0.0469	RT1.079_MZ313.23682	0.0469
RT4.616_MZ183.96486	0.0469	RT1.073_MZ457.36689	0.0469	RT4.943_MZ132.10176	0.0469
RT7.782_MZ619.15384	0.0469	RT8.132_MZ159.07627	0.0469	RT1.038_MZ369.35096	0.0469
RT4.47_MZ174.12351	0.0469	RT1.045_MZ420.40404	0.0469	RT1.419_MZ373.33334	0.0469
RT1.419_MZ444.3678	0.0469	RT1.254_MZ286.34022	0.0469	RT2.383_MZ83.08543	0.0469
RT1.301_MZ193.14312	0.0469	RT1.204_MZ323.18474	0.0469	RT1.546_MZ287.2327	0.0469
RT2.464_MZ139.05009	0.0469	RT1.068_MZ295.22627	0.0469	RT1.074_MZ297.2419	0.0469
RT1.315_MZ298.30998	0.0469	RT1.037_MZ413.37309	0.0469	RT1.103_MZ315.25251	0.0469
RT4.61_MZ183.96487	0.0469				

C HSST3 Significant Feature Lists

C.1 HSST3 - Friedman (Global) (352 features)

Compound	p-val	Compound	p-val	Compound	p-val
RT13.325_MZ300.32532	0.0009	RT12.912_MZ316.28387	0.0021	RT17.085_MZ165.99057	0.0021
RT12.34_MZ281.21046	0.0021	RT13.169_MZ300.32528	0.0021	RT14.024_MZ330.29941	0.0021
RT16.967_MZ799.48856	0.0021	RT9.401_MZ290.23188	0.0021	RT13.218_MZ330.29944	0.0021
RT12.42_MZ302.26828	0.0021	RT12.387_MZ284.2578	0.0021	RT1.855_MZ170.09208	0.0021
RT0.776_MZ216.12022	0.0038	RT12.684_MZ236.0953	0.0038	RT14.766_MZ428.37237	0.0038
RT0.794_MZ85.07643	0.0038	RT13.238_MZ444.36725	0.0038	RT0.751_MZ221.04508	0.0038
RT13.078_MZ225.18445	0.0038	RT4.643_MZ287.14961	0.0038	RT3.107_MZ194.1033	0.0038
RT15.938_MZ328.28377	0.0038	RT0.812_MZ141.10197	0.0038	RT16.967_MZ803.52228	0.0038
RT4.032_MZ129.0545	0.0038	RT0.684_MZ203.01473	0.0038	RT5.737_MZ276.17993	0.0038
RT16.961_MZ406.29743	0.0038	RT16.96_MZ777.50671	0.0038	RT13.136_MZ359.1303	0.0038
RT0.667_MZ145.04686	0.0051	RT0.667_MZ144.10164	0.0051	RT12.993_MZ334.2943	0.0051
RT13.811_MZ189.16337	0.0051	RT0.751_MZ199.06311	0.0051	RT13.498_MZ140.0373	0.0051
RT0.817_MZ182.09204	0.0051	RT0.773_MZ165.1132	0.0051	RT13.797_MZ69.07043	0.0051
RT0.772_MZ140.04521	0.0051	RT16.969_MZ404.28165	0.0051	RT15.815_MZ179.99655	0.0051
RT0.119_MZ472.2154	0.0051	RT0.657_MZ175.05738	0.0051	RT0.637_MZ126.06607	0.0051

Compound	p-val	Compound	p-val	Compound	p-val
RT16.986_MZ284.33035	0.0051	RT13.236_MZ304.28392	0.0051	RT5.347_MZ315.14445	0.0051
RT0.778_MZ140.0816	0.0051	RT14.017_MZ302.30461	0.0058	RT11.902_MZ352.28364	0.0058
RT13.855_MZ282.27848	0.0058	RT13.411_MZ179.0733	0.0058	RT0.811_MZ154.0972	0.0058
RT13.506_MZ360.1505	0.0058	RT15.21_MZ329.27893	0.0058	RT14.494_MZ263.19994	0.0058
RT0.678_MZ114.06626	0.0058	RT0.821_MZ254.16047	0.0058	RT6.405_MZ277.11771	0.0058
RT7.669_MZ185.11447	0.0058	RT16.957_MZ755.52495	0.0058	RT14.496_MZ270.31483	0.0119
RT17.084_MZ88.02347	0.0119	RT16.956_MZ781.54064	0.0119	RT16.974_MZ438.2039	0.0119
RT16.975_MZ422.22982	0.0119	RT0.78_MZ146.10526	0.0119	RT15.271_MZ317.20775	0.0119
RT12.863_MZ298.30981	0.0119	RT11.707_MZ288.2526	0.0119	RT0.55_MZ116.07067	0.0119
RT13.787_MZ343.25823	0.0119	RT16.979_MZ490.2173	0.0119	RT16.98_MZ558.20458	0.0119
RT15.612_MZ546.39877	0.0119	RT10.845_MZ337.24522	0.0119	RT12.532_MZ300.32533	0.0119
RT13.459_MZ256.19004	0.0119	RT14.933_MZ385.37767	0.0119	RT0.825_MZ102.10282	0.0119
RT16.032_MZ560.38118	0.0119	RT0.128_MZ496.41977	0.0119	RT6.945_MZ91.05455	0.0119
RT13.784_MZ239.19996	0.0119	RT12.92_MZ298.2734	0.0119	RT0.815_MZ147.11249	0.0119
RT0.689_MZ180.02681	0.0119	RT11.7_MZ263.16358	0.0119	RT0.119_MZ504.31906	0.0119
RT12.897_MZ300.32536	0.0119	RT13.419_MZ333.14879	0.0119	RT11.738_MZ318.26321	0.0119
RT0.676_MZ110.97545	0.0119	RT0.118_MZ430.16845	0.0119	RT0.654_MZ162.11213	0.0119
RT0.815_MZ123.05525	0.0119	RT13.43_MZ207.10457	0.0119	RT0.126_MZ604.40766	0.0119
RT15.626_MZ182.98477	0.0119	RT13.4_MZ287.20326	0.0119	RT13.56_MZ329.31541	0.0119
RT14.473_MZ281.21047	0.0119	RT13.712_MZ357.31037	0.0119	RT0.815_MZ171.07628	0.0119
RT0.127_MZ524.451	0.0119	RT13.428_MZ221.12017	0.0119	RT3.208_MZ103.05443	0.0119
RT0.813_MZ136.06155	0.0183	RT15.984_MZ345.25608	0.0183	RT15.601_MZ590.42472	0.0183
RT13.843_MZ291.25221	0.0183	RT16.975_MZ701.40731	0.0183	RT13.494_MZ322.20372	0.0183
RT0.719_MZ203.14992	0.0183	RT0.125_MZ649.46484	0.0183	RT3.034_MZ182.09208	0.0183
RT5.994_MZ472.27428	0.0183	RT10.261_MZ511.27832	0.0183	RT0.796_MZ88.02352	0.0183
RT16.358_MZ496.41966	0.0183	RT13.4_MZ328.35637	0.0183	RT0.815_MZ137.04557	0.0183
RT0.821_MZ144.10169	0.0183	RT0.685_MZ181.03284	0.0183	RT12.797_MZ300.32534	0.0183
RT12.832_MZ348.27356	0.0183	RT0.816_MZ165.05435	0.0183	RT14.419_MZ298.34613	0.0183
RT14.683_MZ286.30978	0.0183	RT0.55_MZ133.097	0.0183	RT0.816_MZ119.04917	0.0183
RT16.334_MZ408.36723	0.0183	RT0.795_MZ219.11557	0.0183	RT5.555_MZ243.13348	0.0183
RT9.421_MZ353.13108	0.0183	RT0.815_MZ158.02416	0.0183	RT12.431_MZ300.32533	0.0183
RT0.815_MZ153.04043	0.0183	RT17.075_MZ384.18688	0.0183	RT14.181_MZ297.17312	0.0183
RT16.358_MZ256.26279	0.0183	RT0.123_MZ378.26628	0.0183	RT12.509_MZ289.18262	0.0183
RT0.818_MZ178.09715	0.0183	RT13.835_MZ349.1647	0.0183	RT12.932_MZ262.22905	0.0183
RT1.907_MZ182.08087	0.0183	RT0.117_MZ116.97196	0.0183	RT13.291_MZ358.26911	0.0183
RT0.816_MZ147.04379	0.0183	RT12.049_MZ433.24227	0.0183	RT0.819_MZ127.05013	0.0183
RT8.822_MZ516.33698	0.0183	RT12.682_MZ289.13796	0.0183	RT0.815_MZ195.11249	0.0183
RT14.108_MZ227.19997	0.0183	RT0.818_MZ194.10317	0.0183	RT15.032_MZ298.34602	0.0183
RT14.371_MZ309.20301	0.0183	RT7.99_MZ125.09603	0.0183	RT10.843_MZ212.20043	0.0183
RT0.816_MZ136.0755	0.0183	RT14.477_MZ249.2207	0.0183	RT0.126_MZ416.21272	0.0183
RT0.811_MZ101.03478	0.0183	RT3.33_MZ247.11465	0.0183	RT1.9_MZ165.05439	0.0183
RT12.669_MZ277.1405	0.0183	RT13.837_MZ388.23531	0.0183	RT13.014_MZ254.24722	0.0183
RT15.201_MZ179.17902	0.0183	RT8.347_MZ184.16923	0.0183	RT0.816_MZ182.08087	0.0183
RT6.192_MZ212.11781	0.0183	RT2.879_MZ83.08595	0.0211	RT0.84_MZ120.0808	0.0211
RT0.772_MZ150.10237	0.0211	RT0.713_MZ116.07063	0.0211	RT4.9_MZ423.21902	0.0211
RT13.074_MZ430.22153	0.0211	RT13.19_MZ317.175	0.0211	RT4.302_MZ229.11783	0.0211
RT12.016_MZ242.24729	0.0211	RT4.262_MZ205.0968	0.0211	RT0.816_MZ123.04402	0.0211
RT12.493_MZ299.30498	0.0211	RT5.382_MZ423.21904	0.0211	RT0.821_MZ131.04139	0.0211
RT14.215_MZ350.3157	0.0211	RT5.841_MZ243.13344	0.0211	RT0.814_MZ99.09194	0.0211
RT13.851_MZ181.1583	0.0211	RT0.817_MZ86.09682	0.0211	RT6.543_MZ446.27696	0.0211

Compound	p-val	Compound	p-val	Compound	p-val
RT0.637_MZ120.0655	0.0211	RT2.034_MZ86.09683	0.0211	RT0.713_MZ70.06568	0.0211
RT0.119_MZ78.99941	0.0211	RT13.184_MZ179.07327	0.0211	RT0.838_MZ166.08599	0.0211
RT3.207_MZ166.08595	0.0211	RT13.188_MZ193.08899	0.0211	RT0.118_MZ354.93128	0.0211
RT0.658_MZ166.05294	0.0211	RT14.291_MZ327.3361	0.0211	RT17.077_MZ425.21338	0.0211
RT17.061_MZ72.08127	0.0211	RT13.192_MZ221.1202	0.0211	RT0.817_MZ132.10173	0.0211
RT13.2_MZ374.16631	0.0211	RT0.813_MZ156.06272	0.0211	RT0.778_MZ139.07984	0.0211
RT0.833_MZ103.05445	0.0211	RT3.636_MZ229.11784	0.0211	RT13.193_MZ207.1046	0.0211
RT0.705_MZ127.0501	0.0211	RT4.265_MZ188.07024	0.0211	RT0.817_MZ265.12099	0.0211
RT0.814_MZ70.06568	0.0211	RT14.631_MZ349.27281	0.0211	RT3.207_MZ120.08076	0.0211
RT0.13_MZ400.27186	0.0281	RT16.269_MZ356.31487	0.0281	RT13.911_MZ355.25813	0.0281
RT15.641_MZ154.98987	0.0281	RT13.817_MZ300.3253	0.0281	RT14.316_MZ357.34651	0.0281
RT12.683_MZ329.27942	0.0281	RT0.606_MZ207.97195	0.0281	RT0.545_MZ95.06068	0.0281
RT11.434_MZ277.21559	0.0281	RT0.847_MZ171.02603	0.0281	RT0.117_MZ499.18601	0.0281
RT6.954_MZ181.1008	0.0281	RT15.248_MZ231.2102	0.0281	RT12.926_MZ273.18763	0.0281
RT0.117_MZ99.01398	0.0281	RT0.117_MZ483.1912	0.0281	RT0.116_MZ363.30968	0.0281
RT16.373_MZ416.26643	0.0281	RT15.205_MZ107.0704	0.0281	RT0.127_MZ532.3504	0.0281
RT11.02_MZ302.19545	0.0281	RT6.964_MZ106.06531	0.0281	RT15.726_MZ350.28914	0.0281
RT4.362_MZ196.13284	0.0281	RT15.236_MZ207.17391	0.0281	RT0.821_MZ230.13824	0.0281
RT12.404_MZ137.09587	0.0281	RT5.717_MZ169.07574	0.0281	RT0.812_MZ130.09735	0.0281
RT6.622_MZ197.07803	0.0281	RT0.67_MZ160.09653	0.0498	RT0.118_MZ459.19644	0.0498
RT0.765_MZ277.0888	0.0498	RT13.83_MZ264.25271	0.0498	RT14.03_MZ228.26806	0.0498
RT8.799_MZ261.11156	0.0498	RT12.561_MZ444.29796	0.0498	RT14.731_MZ390.32054	0.0498
RT13.359_MZ388.30472	0.0498	RT14.569_MZ402.35666	0.0498	RT3.121_MZ193.08639	0.0498
RT0.118_MZ468.38862	0.0498	RT0.607_MZ189.13417	0.0498	RT0.638_MZ313.91823	0.0498
RT14.68_MZ284.33008	0.0498	RT12.848_MZ334.29438	0.0498	RT13.958_MZ416.33598	0.0498
RT0.128_MZ516.35558	0.0498	RT16.897_MZ435.36675	0.0498	RT0.127_MZ480.42482	0.0498
RT0.12_MZ427.2716	0.0498	RT0.128_MZ576.37644	0.0498	RT12.686_MZ771.45435	0.0498
RT16.903_MZ364.34106	0.0498	RT0.117_MZ431.1651	0.0498	RT0.668_MZ115.08665	0.0498
RT0.77_MZ187.06853	0.0498	RT0.118_MZ527.2174	0.0498	RT0.758_MZ275.99205	0.0498
RT15.606_MZ284.33034	0.0498	RT0.668_MZ198.9174	0.0498	RT7.129_MZ238.10689	0.0498
RT14.888_MZ298.34605	0.0498	RT0.124_MZ207.01716	0.0498	RT9.767_MZ323.12058	0.0498
RT0.127_MZ560.38156	0.0498	RT13.68_MZ220.22664	0.0498	RT13.885_MZ326.34066	0.0498
RT15.093_MZ430.3879	0.0498	RT0.814_MZ166.07203	0.0498	RT12.146_MZ283.18971	0.0498
RT13.5_MZ443.38324	0.0498	RT0.117_MZ319.28366	0.0498	RT1.638_MZ121.03963	0.0498
RT0.127_MZ533.38183	0.0498	RT16.958_MZ363.19541	0.0498	RT16.124_MZ81.04515	0.0498
RT0.13_MZ303.28871	0.0498	RT0.126_MZ561.4131	0.0498	RT0.677_MZ114.06625	0.0498
RT11.827_MZ294.17278	0.0498	RT12.255_MZ272.29407	0.0498	RT9.897_MZ249.10917	0.0498
RT0.633_MZ164.02905	0.0498	RT14.178_MZ330.31462	0.0498	RT0.799_MZ121.06478	0.0498
RT15.2_MZ197.15319	0.0498	RT15.617_MZ113.96371	0.0498	RT16.186_MZ472.32912	0.0498
RT0.817_MZ185.16452	0.0498	RT16.2_MZ297.23926	0.0498	RT0.119_MZ347.3148	0.0498
RT11.864_MZ241.14285	0.0498	RT0.118_MZ383.24532	0.0498	RT15.926_MZ415.04143	0.0498
RT15.573_MZ678.47679	0.0498	RT15.804_MZ228.23165	0.0498	RT0.816_MZ172.11891	0.0498
RT13.734_MZ340.25077	0.0498	RT16.901_MZ887.58829	0.0498	RT0.118_MZ477.1859	0.0498
RT16.893_MZ367.33496	0.0498	RT0.764_MZ97.03992	0.0498	RT15.794_MZ680.47834	0.0498
RT14.624_MZ291.19808	0.0498	RT11.874_MZ281.13524	0.0498	RT13.373_MZ329.2426	0.0498
RT14.617_MZ400.34128	0.0498	RT13.375_MZ147.11654	0.0498	RT13.796_MZ397.18234	0.0498
RT16.504_MZ383.36209	0.0498	RT13.852_MZ328.35642	0.0498	RT0.714_MZ146.11728	0.0498
RT0.127_MZ452.39361	0.0498	RT17.083_MZ90.50721	0.0498	RT14.29_MZ360.17955	0.0498
RT10.725_MZ238.17962	0.0498	RT14.145_MZ235.20509	0.0498	RT4.549_MZ358.24274	0.0498
RT0.815_MZ146.08093	0.0498	RT12.538_MZ271.2737	0.0498	RT13.797_MZ356.15575	0.0498

Compound	p-val	Compound	p-val	Compound	p-val
RT14.365_MZ299.21868	0.0498	RT16.486_MZ684.56002	0.0498	RT16.861_MZ576.37607	0.0498
RT13.328_MZ533.29208	0.0498	RT10.042_MZ242.24724	0.0498	RT16.36_MZ496.41947	0.0498
RT13.424_MZ589.32051	0.0498				

C.2 HSST3 - Before vs During (148 features)

Compound	p-val	Compound	p-val	Compound	p-val
RT13.325_MZ300.32532	0.0156	RT14.933_MZ385.37767	0.0156	RT0.825_MZ102.10282	0.0156
RT13.291_MZ358.26911	0.0156	RT0.811_MZ101.03478	0.0156	RT5.994_MZ472.27428	0.0156
RT0.125_MZ649.46484	0.0156	RT12.431_MZ300.32533	0.0156	RT16.334_MZ408.36723	0.0156
RT12.832_MZ348.27356	0.0156	RT12.797_MZ300.32534	0.0156	RT14.291_MZ327.3361	0.0156
RT13.074_MZ430.22153	0.0156	RT6.543_MZ446.27696	0.0156	RT0.127_MZ532.3504	0.0156
RT0.117_MZ483.1912	0.0156	RT0.116_MZ363.30968	0.0156	RT0.13_MZ400.27186	0.0156
RT0.117_MZ499.18601	0.0156	RT0.118_MZ383.24532	0.0156	RT0.119_MZ347.3148	0.0156
RT16.186_MZ472.32912	0.0156	RT16.861_MZ576.37607	0.0156	RT14.617_MZ400.34128	0.0156
RT0.12_MZ427.2716	0.0156	RT0.118_MZ468.38862	0.0156	RT0.128_MZ576.37644	0.0156
RT0.118_MZ459.19644	0.0156	RT0.118_MZ527.2174	0.0156	RT14.789_MZ543.43543	0.0156
RT12.532_MZ300.32533	0.0156	RT0.128_MZ496.41977	0.0156	RT0.13_MZ303.28871	0.0156
RT12.863_MZ298.30981	0.0156	RT17.085_MZ165.99057	0.0156	RT14.496_MZ270.31483	0.0156
RT16.032_MZ560.38118	0.0156	RT13.811_MZ189.16337	0.0156	RT16.967_MZ799.48856	0.0156
RT0.119_MZ504.31906	0.0156	RT0.127_MZ524.451	0.0156	RT12.897_MZ300.32536	0.0156
RT0.126_MZ604.40766	0.0156	RT0.118_MZ430.16845	0.0156	RT11.738_MZ318.26321	0.0156
RT14.75_MZ255.23103	0.0312	RT0.129_MZ444.298	0.0312	RT0.131_MZ320.31524	0.0312
RT4.132_MZ297.02979	0.0312	RT0.128_MZ516.35558	0.0312	RT0.119_MZ482.18733	0.0312
RT0.814_MZ79.02164	0.0312	RT0.129_MZ284.33047	0.0312	RT0.117_MZ431.1651	0.0312
RT14.888_MZ298.34605	0.0312	RT15.93_MZ338.34067	0.0312	RT17.083_MZ90.01995	0.0312
RT17.078_MZ207.01699	0.0312	RT0.691_MZ140.06793	0.0312	RT17.08_MZ81.01472	0.0312
RT0.127_MZ392.37241	0.0312	RT0.559_MZ216.92228	0.0312	RT12.597_MZ288.28895	0.0312
RT0.776_MZ176.1278	0.0312	RT15.623_MZ163.99016	0.0312	RT12.429_MZ224.06251	0.0312
RT0.119_MZ424.36253	0.0312	RT15.606_MZ284.33034	0.0312	RT12.704_MZ262.2007	0.0312
RT0.118_MZ461.19707	0.0312	RT14.365_MZ299.21868	0.0312	RT0.78_MZ146.10526	0.0312
RT0.124_MZ167.97435	0.0312	RT7.99_MZ125.09603	0.0312	RT13.1_MZ242.28361	0.0312
RT14.473_MZ281.21047	0.0312	RT15.601_MZ590.42472	0.0312	RT0.676_MZ110.97545	0.0312
RT16.989_MZ380.33614	0.0312	RT0.138_MZ129.0406	0.0312	RT17.086_MZ117.95947	0.0312
RT15.236_MZ207.17391	0.0312	RT0.815_MZ146.08093	0.0312	RT13.911_MZ355.25813	0.0312
RT11.902_MZ352.28364	0.0312	RT16.901_MZ887.58829	0.0312	RT0.119_MZ391.34085	0.0312
RT0.119_MZ480.19188	0.0312	RT12.912_MZ316.28387	0.0312	RT7.669_MZ185.11447	0.0312
RT15.194_MZ167.14271	0.0469	RT0.815_MZ135.00269	0.0469	RT0.129_MZ488.32415	0.0469
RT14.159_MZ325.32048	0.0469	RT0.127_MZ449.25354	0.0469	RT0.127_MZ436.39871	0.0469
RT16.731_MZ517.36997	0.0469	RT12.389_MZ316.32029	0.0469	RT15.199_MZ129.09085	0.0469
RT16.966_MZ647.33946	0.0469	RT12.258_MZ160.09365	0.0469	RT17.085_MZ98.01381	0.0469
RT0.129_MZ425.21358	0.0469	RT15.201_MZ322.27323	0.0469	RT14.139_MZ318.23961	0.0469
RT14.968_MZ279.23115	0.0469	RT15.881_MZ328.21433	0.0469	RT17.079_MZ166.9939	0.0469
RT15.271_MZ317.20775	0.0469	RT14.343_MZ263.19996	0.0469	RT12.34_MZ281.21046	0.0469
RT0.127_MZ452.39361	0.0469	RT17.083_MZ90.50721	0.0469	RT13.328_MZ533.29208	0.0469
RT10.042_MZ242.24724	0.0469	RT0.821_MZ254.16047	0.0469	RT0.545_MZ95.06068	0.0469
RT0.127_MZ480.42482	0.0469	RT13.817_MZ300.3253	0.0469	RT13.855_MZ282.27848	0.0469
RT14.017_MZ302.30461	0.0469	RT15.205_MZ107.0704	0.0469	RT15.21_MZ329.27893	0.0469

Compound	p-val	Compound	p-val	Compound	p-val
RT15.626_MZ182.98477	0.0469	RT14.371_MZ309.20301	0.0469	RT17.084_MZ88.02347	0.0469
RT16.957_MZ755.52495	0.0469	RT15.31_MZ285.24175	0.0469	RT13.169_MZ300.32528	0.0469
RT12.561_MZ444.29796	0.0469	RT0.131_MZ405.22729	0.0469	RT13.672_MZ256.29924	0.0469
RT12.758_MZ157.08571	0.0469	RT13.185_MZ385.37785	0.0469	RT12.387_MZ284.2578	0.0469
RT0.126_MZ605.43884	0.0469	RT0.128_MZ540.44592	0.0469	RT14.024_MZ330.29941	0.0469
RT16.087_MZ354.2994	0.0469	RT0.128_MZ408.36745	0.0469	RT13.218_MZ330.29944	0.0469
RT0.758_MZ275.99205	0.0469	RT0.127_MZ560.38156	0.0469	RT16.903_MZ364.34106	0.0469
RT0.117_MZ319.28366	0.0469	RT9.401_MZ290.23188	0.0469	RT12.57_MZ347.31476	0.0469
RT0.126_MZ421.19179	0.0469				

C.3 HSST3 - During vs After (316 features)

Compound	p-val	Compound	p-val	Compound	p-val
RT13.325_MZ300.32532	0.0156	RT0.813_MZ156.06272	0.0156	RT13.193_MZ207.1046	0.0156
RT0.705_MZ127.0501	0.0156	RT13.184_MZ179.07327	0.0156	RT13.19_MZ317.175	0.0156
RT4.9_MZ423.21902	0.0156	RT12.016_MZ242.24729	0.0156	RT13.192_MZ221.1202	0.0156
RT2.879_MZ83.08595	0.0156	RT5.382_MZ423.21904	0.0156	RT0.814_MZ99.09194	0.0156
RT16.373_MZ416.26643	0.0156	RT6.964_MZ106.06531	0.0156	RT4.362_MZ196.13284	0.0156
RT12.404_MZ137.09587	0.0156	RT6.954_MZ181.1008	0.0156	RT12.493_MZ299.30498	0.0156
RT0.117_MZ99.01398	0.0156	RT13.2_MZ374.16631	0.0156	RT9.421_MZ353.13108	0.0156
RT12.932_MZ262.22905	0.0156	RT15.032_MZ298.34602	0.0156	RT12.049_MZ433.24227	0.0156
RT8.822_MZ516.33698	0.0156	RT12.682_MZ289.13796	0.0156	RT7.99_MZ125.09603	0.0156
RT8.347_MZ184.16923	0.0156	RT13.188_MZ193.08899	0.0156	RT15.201_MZ179.17902	0.0156
RT0.123_MZ378.26628	0.0156	RT16.358_MZ496.41966	0.0156	RT0.796_MZ88.02352	0.0156
RT13.4_MZ328.35637	0.0156	RT15.984_MZ345.25608	0.0156	RT15.601_MZ590.42472	0.0156
RT0.815_MZ158.02416	0.0156	RT3.33_MZ247.11465	0.0156	RT0.818_MZ178.09715	0.0156
RT12.683_MZ329.27942	0.0156	RT0.817_MZ185.16452	0.0156	RT3.999_MZ358.24267	0.0156
RT14.055_MZ319.22943	0.0156	RT4.006_MZ314.21659	0.0156	RT16.037_MZ255.23123	0.0156
RT16.062_MZ445.19662	0.0156	RT15.685_MZ242.2472	0.0156	RT5.173_MZ357.21101	0.0156
RT12.725_MZ242.24727	0.0156	RT0.644_MZ308.76166	0.0156	RT12.683_MZ542.2524	0.0156
RT12.754_MZ385.30507	0.0156	RT6.108_MZ415.25278	0.0156	RT11.584_MZ186.22128	0.0156
RT15.731_MZ328.28369	0.0156	RT15.832_MZ512.41457	0.0156	RT17.078_MZ161.96914	0.0156
RT5.166_MZ402.26882	0.0156	RT16.2_MZ297.23926	0.0156	RT3.582_MZ247.11468	0.0156
RT13.833_MZ309.20317	0.0156	RT14.145_MZ235.20509	0.0156	RT0.638_MZ313.91823	0.0156
RT0.607_MZ189.13417	0.0156	RT0.67_MZ160.09653	0.0156	RT12.686_MZ771.45435	0.0156
RT12.146_MZ283.18971	0.0156	RT0.611_MZ103.52065	0.0156	RT0.451_MZ116.10705	0.0156
RT0.723_MZ251.12315	0.0156	RT15.186_MZ255.23115	0.0156	RT16.411_MZ816.63841	0.0156
RT16.964_MZ401.15123	0.0156	RT4.548_MZ330.21148	0.0156	RT0.759_MZ97.02868	0.0156
RT12.684_MZ511.31062	0.0156	RT16.89_MZ488.3238	0.0156	RT16.858_MZ385.2703	0.0156
RT12.509_MZ289.18262	0.0156	RT16.36_MZ496.41947	0.0156	RT12.993_MZ334.2943	0.0156
RT0.657_MZ175.05738	0.0156	RT17.085_MZ165.99057	0.0156	RT12.34_MZ281.21046	0.0156
RT0.119_MZ472.2154	0.0156	RT13.169_MZ300.32528	0.0156	RT14.024_MZ330.29941	0.0156
RT16.969_MZ404.28165	0.0156	RT15.815_MZ179.99655	0.0156	RT13.797_MZ69.07043	0.0156
RT0.667_MZ145.04686	0.0156	RT1.855_MZ170.09208	0.0156	RT0.772_MZ140.04521	0.0156
RT0.667_MZ144.10164	0.0156	RT13.459_MZ256.19004	0.0156	RT0.773_MZ165.1132	0.0156
RT13.498_MZ140.0373	0.0156	RT0.817_MZ182.09204	0.0156	RT0.751_MZ199.06311	0.0156
RT0.821_MZ254.16047	0.0156	RT0.637_MZ126.06607	0.0156	RT0.678_MZ114.06626	0.0156
RT16.986_MZ284.33035	0.0156	RT16.96_MZ777.50671	0.0156	RT3.107_MZ194.1033	0.0156

Compound	p-val	Compound	p-val	Compound	p-val
RT13.136_MZ359.1303	0.0156	RT15.938_MZ328.28377	0.0156	RT0.812_MZ141.10197	0.0156
RT4.643_MZ287.14961	0.0156	RT13.078_MZ225.18445	0.0156	RT0.751_MZ221.04508	0.0156
RT13.238_MZ444.36725	0.0156	RT0.794_MZ85.07643	0.0156	RT16.967_MZ803.52228	0.0156
RT14.766_MZ428.37237	0.0156	RT4.032_MZ129.0545	0.0156	RT12.684_MZ236.0953	0.0156
RT0.778_MZ140.0816	0.0156	RT0.684_MZ203.01473	0.0156	RT5.737_MZ276.17993	0.0156
RT16.961_MZ406.29743	0.0156	RT5.347_MZ315.14445	0.0156	RT13.236_MZ304.28392	0.0156
RT12.912_MZ316.28387	0.0156	RT9.401_MZ290.23188	0.0156	RT0.776_MZ216.12022	0.0156
RT0.78_MZ146.10526	0.0156	RT13.218_MZ330.29944	0.0156	RT12.42_MZ302.26828	0.0156
RT0.815_MZ123.05525	0.0156	RT15.612_MZ546.39877	0.0156	RT6.945_MZ91.05455	0.0156
RT0.676_MZ110.97545	0.0156	RT13.56_MZ329.31541	0.0156	RT0.654_MZ162.11213	0.0156
RT12.387_MZ284.2578	0.0156	RT0.755_MZ180.08628	0.0312	RT13.709_MZ741.6444	0.0312
RT0.775_MZ120.06551	0.0312	RT0.815_MZ180.08629	0.0312	RT0.701_MZ99.04248	0.0312
RT13.567_MZ318.29958	0.0312	RT0.816_MZ235.11846	0.0312	RT0.758_MZ291.06847	0.0312
RT0.524_MZ69.98357	0.0312	RT12.68_MZ208.06407	0.0312	RT12.983_MZ491.50522	0.0312
RT15.799_MZ663.45173	0.0312	RT12.403_MZ325.16769	0.0312	RT0.155_MZ171.99261	0.0312
RT16.125_MZ265.25187	0.0312	RT11.865_MZ259.15343	0.0312	RT16.977_MZ516.23293	0.0312
RT13.354_MZ272.29407	0.0312	RT4.954_MZ275.14585	0.0312	RT15.815_MZ369.11462	0.0312
RT16.493_MZ424.36194	0.0312	RT16.126_MZ282.27833	0.0312	RT13.933_MZ368.31495	0.0312
RT15.86_MZ413.22472	0.0312	RT12.683_MZ527.28798	0.0312	RT12.704_MZ262.2007	0.0312
RT0.628_MZ306.76468	0.0312	RT16.369_MZ336.30981	0.0312	RT0.758_MZ411.09746	0.0312
RT0.675_MZ74.07179	0.0312	RT0.766_MZ132.06537	0.0312	RT0.669_MZ95.02428	0.0312
RT0.821_MZ83.08597	0.0312	RT0.72_MZ272.87076	0.0312	RT12.688_MZ533.30527	0.0312
RT15.586_MZ634.45074	0.0312	RT12.409_MZ332.27866	0.0312	RT0.755_MZ388.18053	0.0312
RT0.112_MZ98.07155	0.0312	RT8.346_MZ287.23214	0.0312	RT12.128_MZ207.17399	0.0312
RT14.289_MZ149.05946	0.0312	RT10.904_MZ254.28363	0.0312	RT16.978_MZ584.22014	0.0312
RT13.71_MZ344.20932	0.0312	RT14.104_MZ179.07327	0.0312	RT0.814_MZ174.03848	0.0312
RT17.077_MZ174.95631	0.0312	RT11.954_MZ226.21604	0.0312	RT16.904_MZ391.34057	0.0312
RT0.125_MZ404.28188	0.0312	RT3.579_MZ270.19047	0.0312	RT2.882_MZ100.11227	0.0312
RT16.36_MZ443.26616	0.0312	RT0.821_MZ100.11229	0.0312	RT16.967_MZ799.48856	0.0312
RT15.201_MZ165.16346	0.0312	RT6.932_MZ198.12735	0.0312	RT14.323_MZ227.2	0.0312
RT16.975_MZ422.22982	0.0312	RT13.014_MZ254.24722	0.0312	RT0.545_MZ95.06068	0.0312
RT15.248_MZ231.2102	0.0312	RT16.269_MZ356.31487	0.0312	RT13.817_MZ300.3253	0.0312
RT16.974_MZ438.2039	0.0312	RT0.126_MZ416.21272	0.0312	RT10.843_MZ212.20043	0.0312
RT6.543_MZ446.27696	0.0312	RT16.956_MZ781.54064	0.0312	RT16.358_MZ256.26279	0.0312
RT5.994_MZ472.27428	0.0312	RT15.804_MZ228.23165	0.0312	RT14.494_MZ263.19994	0.0312
RT12.92_MZ298.2734	0.0312	RT0.811_MZ154.0972	0.0312	RT4.378_MZ395.16704	0.0312
RT11.902_MZ352.28364	0.0312	RT14.017_MZ302.30461	0.0312	RT12.897_MZ300.32536	0.0312
RT17.061_MZ72.08127	0.0312	RT0.813_MZ136.06155	0.0312	RT14.683_MZ286.30978	0.0312
RT13.4_MZ287.20326	0.0312	RT13.712_MZ357.31037	0.0312	RT0.815_MZ137.04557	0.0312
RT16.957_MZ755.52495	0.0312	RT0.764_MZ97.03992	0.0312	RT10.261_MZ511.27832	0.0312
RT16.124_MZ81.04515	0.0312	RT16.979_MZ490.2173	0.0312	RT16.629_MZ284.29399	0.0312
RT13.784_MZ239.19996	0.0312	RT14.108_MZ227.19997	0.0312	RT13.5_MZ443.38324	0.0312
RT16.98_MZ558.20458	0.0312	RT14.569_MZ402.35666	0.0312	RT14.68_MZ284.33008	0.0312
RT0.755_MZ401.06862	0.0312	RT13.852_MZ328.35642	0.0312	RT11.707_MZ288.2526	0.0312
RT16.97_MZ385.17722	0.0312	RT0.818_MZ138.09112	0.0312	RT0.818_MZ123.09163	0.0312
RT0.812_MZ97.07632	0.0312	RT16.034_MZ256.26276	0.0312	RT10.843_MZ315.26341	0.0312
RT10.845_MZ337.24522	0.0312	RT0.691_MZ140.06793	0.0312	RT13.787_MZ343.25823	0.0469
RT16.537_MZ468.38818	0.0469	RT0.814_MZ152.05642	0.0469	RT13.494_MZ322.20372	0.0469
RT16.975_MZ701.40731	0.0469	RT12.542_MZ329.279	0.0469	RT15.831_MZ235.20505	0.0469
RT5.375_MZ401.23722	0.0469	RT16.022_MZ368.31494	0.0469	RT16.436_MZ772.6123	0.0469

Compound	p-val	Compound	p-val	Compound	p-val
RT12.664_MZ285.28924	0.0469	RT16.959_MZ359.10431	0.0469	RT15.844_MZ407.3356	0.0469
RT10.96_MZ313.16129	0.0469	RT6.192_MZ212.11781	0.0469	RT17.086_MZ232.88964	0.0469
RT16.973_MZ411.11762	0.0469	RT17.08_MZ209.01244	0.0469	RT12.669_MZ277.1405	0.0469
RT14.961_MZ170.1172	0.0469	RT0.779_MZ145.1081	0.0469	RT0.818_MZ194.10317	0.0469
RT16.041_MZ477.16857	0.0469	RT16.488_MZ592.37091	0.0469	RT3.034_MZ182.09208	0.0469
RT12.645_MZ301.28414	0.0469	RT12.687_MZ267.15601	0.0469	RT16.22_MZ111.11691	0.0469
RT0.799_MZ121.06478	0.0469	RT12.255_MZ272.29407	0.0469	RT16.2_MZ409.20005	0.0469
RT11.827_MZ294.17278	0.0469	RT14.365_MZ299.21868	0.0469	RT11.874_MZ281.13524	0.0469
RT8.799_MZ261.11156	0.0469	RT13.958_MZ416.33598	0.0469	RT11.864_MZ241.14285	0.0469
RT0.814_MZ166.07203	0.0469	RT13.68_MZ220.22664	0.0469	RT17.078_MZ207.01699	0.0469
RT0.816_MZ170.09208	0.0469	RT13.27_MZ302.26831	0.0469	RT0.776_MZ176.1278	0.0469
RT15.905_MZ336.30978	0.0469	RT11.193_MZ336.23719	0.0469	RT0.818_MZ215.01576	0.0469
RT15.093_MZ430.3879	0.0469	RT15.926_MZ415.04143	0.0469	RT15.573_MZ678.47679	0.0469
RT17.079_MZ166.9939	0.0469	RT17.082_MZ140.98523	0.0469	RT13.428_MZ221.12017	0.0469
RT14.419_MZ298.34613	0.0469	RT13.43_MZ207.10457	0.0469	RT0.755_MZ765.17845	0.0469
RT13.411_MZ179.0733	0.0469	RT16.421_MZ425.17725	0.0469	RT6.164_MZ504.33679	0.0469
RT13.034_MZ543.50828	0.0469	RT0.812_MZ158.08091	0.0469	RT15.002_MZ350.23489	0.0469
RT14.215_MZ350.3157	0.0469	RT0.812_MZ130.09735	0.0469	RT17.083_MZ125.9721	0.0469
RT4.747_MZ155.15398	0.0469	RT0.62_MZ120.10188	0.0469	RT13.106_MZ343.29474	0.0469
RT14.176_MZ300.3253	0.0469	RT11.614_MZ383.43503	0.0469	RT5.904_MZ402.25091	0.0469
RT12.532_MZ300.32533	0.0469				

C.4 HSST3 - Before vs After (364 features)

Compound	p-val	Compound	p-val	Compound	p-val
RT13.325_MZ300.32532	0.0156	RT13.843_MZ291.25221	0.0156	RT10.261_MZ511.27832	0.0156
RT0.815_MZ137.04557	0.0156	RT0.816_MZ119.04917	0.0156	RT0.685_MZ181.03284	0.0156
RT0.815_MZ153.04043	0.0156	RT0.821_MZ144.10169	0.0156	RT5.555_MZ243.13348	0.0156
RT0.795_MZ219.11557	0.0156	RT16.975_MZ701.40731	0.0156	RT17.075_MZ384.18688	0.0156
RT14.683_MZ286.30978	0.0156	RT14.419_MZ298.34613	0.0156	RT0.816_MZ165.05435	0.0156
RT0.119_MZ78.99941	0.0156	RT3.207_MZ166.08595	0.0156	RT0.118_MZ354.93128	0.0156
RT0.658_MZ166.05294	0.0156	RT17.077_MZ425.21338	0.0156	RT17.061_MZ72.08127	0.0156
RT14.181_MZ297.17312	0.0156	RT0.817_MZ132.10173	0.0156	RT0.813_MZ136.06155	0.0156
RT3.034_MZ182.09208	0.0156	RT0.816_MZ147.04379	0.0156	RT0.819_MZ127.05013	0.0156
RT0.815_MZ195.11249	0.0156	RT14.108_MZ227.19997	0.0156	RT0.818_MZ194.10317	0.0156
RT14.371_MZ309.20301	0.0156	RT6.192_MZ212.11781	0.0156	RT13.014_MZ254.24722	0.0156
RT13.837_MZ388.23531	0.0156	RT13.494_MZ322.20372	0.0156	RT12.669_MZ277.1405	0.0156
RT0.126_MZ416.21272	0.0156	RT14.477_MZ249.2207	0.0156	RT1.9_MZ165.05439	0.0156
RT0.816_MZ136.0755	0.0156	RT10.843_MZ212.20043	0.0156	RT1.907_MZ182.08087	0.0156
RT16.358_MZ256.26279	0.0156	RT0.55_MZ133.097	0.0156	RT0.719_MZ203.14992	0.0156
RT0.816_MZ182.08087	0.0156	RT0.117_MZ116.97196	0.0156	RT0.778_MZ139.07984	0.0156
RT0.814_MZ70.06568	0.0156	RT9.897_MZ249.10917	0.0156	RT14.624_MZ291.19808	0.0156
RT13.797_MZ356.15575	0.0156	RT13.796_MZ397.18234	0.0156	RT0.765_MZ277.0888	0.0156
RT13.83_MZ264.25271	0.0156	RT13.359_MZ388.30472	0.0156	RT0.124_MZ207.01716	0.0156
RT14.996_MZ253.21553	0.0156	RT13.734_MZ340.25077	0.0156	RT0.607_MZ323.88881	0.0156
RT0.127_MZ153.95881	0.0156	RT0.567_MZ191.0763	0.0156	RT0.783_MZ248.13481	0.0156
RT0.819_MZ181.09688	0.0156	RT12.837_MZ357.31025	0.0156	RT15.145_MZ312.3612	0.0156
RT0.607_MZ432.99491	0.0156	RT15.369_MZ340.39255	0.0156	RT6.625_MZ231.05354	0.0156

Compound	p-val	Compound	p-val	Compound	p-val
RT15.461_MZ366.40828	0.0156	RT14.631_MZ349.27281	0.0156	RT0.847_MZ171.02603	0.0156
RT6.622_MZ197.07803	0.0156	RT0.817_MZ265.12099	0.0156	RT4.265_MZ188.07024	0.0156
RT3.207_MZ120.08076	0.0156	RT3.636_MZ229.11784	0.0156	RT0.833_MZ103.05445	0.0156
RT0.838_MZ166.08599	0.0156	RT0.713_MZ70.06568	0.0156	RT0.637_MZ120.0655	0.0156
RT0.713_MZ116.07063	0.0156	RT0.606_MZ207.97195	0.0156	RT0.772_MZ150.10237	0.0156
RT4.302_MZ229.11783	0.0156	RT4.262_MZ205.0968	0.0156	RT0.821_MZ131.04139	0.0156
RT14.215_MZ350.3157	0.0156	RT5.841_MZ243.13344	0.0156	RT0.816_MZ123.04402	0.0156
RT13.851_MZ181.1583	0.0156	RT0.817_MZ86.09682	0.0156	RT11.02_MZ302.19545	0.0156
RT0.84_MZ120.0808	0.0156	RT13.835_MZ349.1647	0.0156	RT2.034_MZ86.09683	0.0156
RT4.032_MZ129.0545	0.0156	RT5.347_MZ315.14445	0.0156	RT13.236_MZ304.28392	0.0156
RT16.986_MZ284.33035	0.0156	RT0.637_MZ126.06607	0.0156	RT0.657_MZ175.05738	0.0156
RT0.119_MZ472.2154	0.0156	RT16.969_MZ404.28165	0.0156	RT15.815_MZ179.99655	0.0156
RT13.797_MZ69.07043	0.0156	RT0.667_MZ145.04686	0.0156	RT0.772_MZ140.04521	0.0156
RT0.667_MZ144.10164	0.0156	RT12.993_MZ334.2943	0.0156	RT13.811_MZ189.16337	0.0156
RT0.773_MZ165.1132	0.0156	RT13.498_MZ140.0373	0.0156	RT0.817_MZ182.09204	0.0156
RT0.751_MZ199.06311	0.0156	RT7.669_MZ185.11447	0.0156	RT6.405_MZ277.11771	0.0156
RT14.494_MZ263.19994	0.0156	RT16.957_MZ755.52495	0.0156	RT13.506_MZ360.1505	0.0156
RT0.778_MZ140.0816	0.0156	RT0.811_MZ154.0972	0.0156	RT12.684_MZ236.0953	0.0156
RT0.794_MZ85.07643	0.0156	RT12.42_MZ302.26828	0.0156	RT13.218_MZ330.29944	0.0156
RT9.401_MZ290.23188	0.0156	RT12.387_MZ284.2578	0.0156	RT1.855_MZ170.09208	0.0156
RT14.024_MZ330.29941	0.0156	RT13.169_MZ300.32528	0.0156	RT12.34_MZ281.21046	0.0156
RT16.967_MZ799.48856	0.0156	RT12.912_MZ316.28387	0.0156	RT16.96_MZ777.50671	0.0156
RT16.961_MZ406.29743	0.0156	RT5.737_MZ276.17993	0.0156	RT0.684_MZ203.01473	0.0156
RT16.967_MZ803.52228	0.0156	RT0.812_MZ141.10197	0.0156	RT15.938_MZ328.28377	0.0156
RT13.136_MZ359.1303	0.0156	RT3.107_MZ194.1033	0.0156	RT4.643_MZ287.14961	0.0156
RT13.078_MZ225.18445	0.0156	RT0.751_MZ221.04508	0.0156	RT13.238_MZ444.36725	0.0156
RT14.766_MZ428.37237	0.0156	RT13.411_MZ179.0733	0.0156	RT0.776_MZ216.12022	0.0156
RT11.902_MZ352.28364	0.0156	RT14.473_MZ281.21047	0.0156	RT13.712_MZ357.31037	0.0156
RT13.4_MZ287.20326	0.0156	RT15.626_MZ182.98477	0.0156	RT12.92_MZ298.2734	0.0156
RT17.084_MZ88.02347	0.0156	RT16.956_MZ781.54064	0.0156	RT16.974_MZ438.2039	0.0156
RT16.975_MZ422.22982	0.0156	RT15.271_MZ317.20775	0.0156	RT11.707_MZ288.2526	0.0156
RT13.855_MZ282.27848	0.0156	RT13.784_MZ239.19996	0.0156	RT16.979_MZ490.2173	0.0156
RT13.787_MZ343.25823	0.0156	RT16.98_MZ558.20458	0.0156	RT10.845_MZ337.24522	0.0156
RT3.208_MZ103.05443	0.0156	RT0.815_MZ171.07628	0.0156	RT0.55_MZ116.07067	0.0156
RT13.43_MZ207.10457	0.0156	RT13.428_MZ221.12017	0.0156	RT15.21_MZ329.27893	0.0156
RT0.689_MZ180.02681	0.0156	RT14.017_MZ302.30461	0.0156	RT11.7_MZ263.16358	0.0156
RT13.419_MZ333.14879	0.0156	RT0.815_MZ147.11249	0.0156	RT15.032_MZ298.34602	0.0312
RT0.698_MZ94.06543	0.0312	RT0.653_MZ210.01673	0.0312	RT4.378_MZ395.16704	0.0312
RT12.832_MZ348.27356	0.0312	RT14.209_MZ345.2061	0.0312	RT0.815_MZ246.14435	0.0312
RT0.82_MZ129.02146	0.0312	RT0.818_MZ178.09715	0.0312	RT0.606_MZ381.84694	0.0312
RT10.843_MZ315.26341	0.0312	RT3.338_MZ205.09679	0.0312	RT0.695_MZ198.98862	0.0312
RT3.773_MZ181.07164	0.0312	RT0.818_MZ138.09112	0.0312	RT16.946_MZ196.86458	0.0312
RT16.629_MZ284.29399	0.0312	RT0.842_MZ314.21655	0.0312	RT15.751_MZ399.06443	0.0312
RT3.992_MZ262.16433	0.0312	RT11.432_MZ295.22614	0.0312	RT12.509_MZ289.18262	0.0312
RT0.788_MZ175.10741	0.0312	RT13.424_MZ589.32051	0.0312	RT15.606_MZ284.33034	0.0312
RT7.129_MZ238.10689	0.0312	RT14.139_MZ167.14271	0.0312	RT13.459_MZ256.19004	0.0312
RT0.118_MZ165.99072	0.0312	RT14.438_MZ369.27389	0.0312	RT10.904_MZ254.28363	0.0312
RT11.614_MZ383.43503	0.0312	RT4.031_MZ242.01044	0.0312	RT16.978_MZ584.22014	0.0312
RT0.617_MZ134.04461	0.0312	RT13.71_MZ344.20932	0.0312	RT0.795_MZ236.14871	0.0312
RT0.814_MZ174.05784	0.0312	RT11.66_MZ300.28906	0.0312	RT13.841_MZ308.27876	0.0312

Compound	p-val	Compound	p-val	Compound	p-val
RT0.808_MZ267.10818	0.0312	RT13.844_MZ247.22619	0.0312	RT12.059_MZ270.2785	0.0312
RT0.829_MZ204.88847	0.0312	RT0.125_MZ404.28188	0.0312	RT0.859_MZ115.95751	0.0312
RT0.819_MZ229.97415	0.0312	RT0.816_MZ235.11846	0.0312	RT12.532_MZ300.32533	0.0312
RT0.82_MZ209.09174	0.0312	RT13.291_MZ358.26911	0.0312	RT15.799_MZ663.45173	0.0312
RT16.977_MZ516.23293	0.0312	RT14.888_MZ298.34605	0.0312	RT12.597_MZ288.28895	0.0312
RT15.722_MZ787.49943	0.0312	RT16.494_MZ524.45079	0.0312	RT6.188_MZ210.10962	0.0312
RT13.68_MZ220.22664	0.0312	RT15.926_MZ415.04143	0.0312	RT15.804_MZ228.23165	0.0312
RT0.696_MZ252.07476	0.0312	RT16.901_MZ887.58829	0.0312	RT3.33_MZ247.11465	0.0312
RT11.434_MZ277.21559	0.0312	RT16.962_MZ669.32109	0.0312	RT0.122_MZ166.98835	0.0312
RT13.817_MZ300.3253	0.0312	RT13.911_MZ355.25813	0.0312	RT11.169_MZ165.1271	0.0312
RT0.604_MZ106.05003	0.0312	RT15.205_MZ107.0704	0.0312	RT6.945_MZ91.05455	0.0312
RT12.897_MZ300.32536	0.0312	RT0.814_MZ99.09194	0.0312	RT0.123_MZ378.26628	0.0312
RT11.738_MZ318.26321	0.0312	RT0.619_MZ178.00835	0.0312	RT0.654_MZ162.11213	0.0312
RT2.879_MZ83.08595	0.0312	RT12.471_MZ294.25346	0.0312	RT13.56_MZ329.31541	0.0312
RT0.705_MZ127.0501	0.0312	RT4.9_MZ423.21902	0.0312	RT13.074_MZ430.22153	0.0312
RT15.984_MZ345.25608	0.0312	RT15.617_MZ113.96371	0.0312	RT0.799_MZ121.06478	0.0312
RT0.118_MZ241.88333	0.0312	RT13.328_MZ533.29208	0.0312	RT4.549_MZ358.24274	0.0312
RT3.121_MZ193.08639	0.0312	RT12.848_MZ334.29438	0.0312	RT14.68_MZ284.33008	0.0312
RT0.821_MZ254.16047	0.0312	RT0.668_MZ115.08665	0.0312	RT15.794_MZ680.47834	0.0312
RT0.764_MZ97.03992	0.0312	RT13.506_MZ344.1857	0.0312	RT16.893_MZ367.33496	0.0312
RT14.933_MZ385.37767	0.0312	RT11.827_MZ294.17278	0.0312	RT13.912_MZ85.0288	0.0312
RT14.95_MZ572.45079	0.0312	RT14.569_MZ402.35666	0.0312	RT11.102_MZ401.21604	0.0469
RT14.674_MZ296.29402	0.0469	RT16.952_MZ445.09424	0.0469	RT8.781_MZ258.2422	0.0469
RT0.715_MZ176.05507	0.0469	RT15.355_MZ382.43948	0.0469	RT0.816_MZ214.08176	0.0469
RT13.883_MZ256.29921	0.0469	RT13.913_MZ387.2133	0.0469	RT0.712_MZ106.05004	0.0469
RT0.56_MZ186.00708	0.0469	RT0.811_MZ129.10217	0.0469	RT15.395_MZ399.17731	0.0469
RT0.835_MZ211.09704	0.0469	RT0.15_MZ108.51766	0.0469	RT14.169_MZ314.30452	0.0469
RT0.816_MZ205.11791	0.0469	RT13.914_MZ719.48966	0.0469	RT11.699_MZ249.18434	0.0469
RT0.799_MZ319.09923	0.0469	RT0.817_MZ117.06593	0.0469	RT10.766_MZ272.29411	0.0469
RT15.814_MZ236.05898	0.0469	RT0.118_MZ238.17968	0.0469	RT1.175_MZ122.08113	0.0469
RT3.836_MZ165.06556	0.0469	RT12.425_MZ223.06316	0.0469	RT0.823_MZ129.06575	0.0469
RT0.825_MZ102.10282	0.0469	RT13.771_MZ257.21044	0.0469	RT0.677_MZ114.06625	0.0469
RT16.504_MZ383.36209	0.0469	RT12.255_MZ272.29407	0.0469	RT0.633_MZ164.02905	0.0469
RT14.178_MZ330.31462	0.0469	RT0.545_MZ95.06068	0.0469	RT16.269_MZ356.31487	0.0469
RT12.926_MZ273.18763	0.0469	RT5.717_MZ169.07574	0.0469	RT5.382_MZ423.21904	0.0469
RT8.799_MZ261.11156	0.0469	RT0.678_MZ114.06626	0.0469	RT9.421_MZ353.13108	0.0469
RT0.815_MZ158.02416	0.0469	RT12.431_MZ300.32533	0.0469	RT0.796_MZ88.02352	0.0469
RT16.358_MZ496.41966	0.0469	RT0.815_MZ123.05525	0.0469	RT0.811_MZ101.03478	0.0469
RT12.863_MZ298.30981	0.0469	RT8.822_MZ516.33698	0.0469	RT12.932_MZ262.22905	0.0469
RT12.797_MZ300.32534	0.0469	RT14.222_MZ388.34102	0.0469	RT0.117_MZ431.1651	0.0469
RT0.742_MZ176.1027	0.0469	RT13.507_MZ227.19998	0.0469	RT10.493_MZ286.23698	0.0469
RT0.119_MZ458.2006	0.0469	RT16.2_MZ409.20005	0.0469	RT10.219_MZ341.38813	0.0469
RT13.626_MZ326.3407	0.0469	RT0.814_MZ113.03456	0.0469	RT14.936_MZ310.34575	0.0469
RT14.343_MZ263.19996	0.0469	RT16.994_MZ460.29265	0.0469	RT9.767_MZ323.12058	0.0469
RT10.441_MZ341.38819	0.0469	RT15.623_MZ163.99016	0.0469	RT15.905_MZ336.30978	0.0469
RT15.86_MZ413.22472	0.0469	RT0.585_MZ84.91164	0.0469	RT0.817_MZ167.99648	0.0469
RT16.034_MZ256.26276	0.0469	RT15.93_MZ338.34067	0.0469	RT11.197_MZ329.13531	0.0469
RT0.816_MZ170.09208	0.0469	RT14.75_MZ255.23103	0.0469	RT10.644_MZ249.14797	0.0469
RT0.652_MZ165.07544	0.0469				